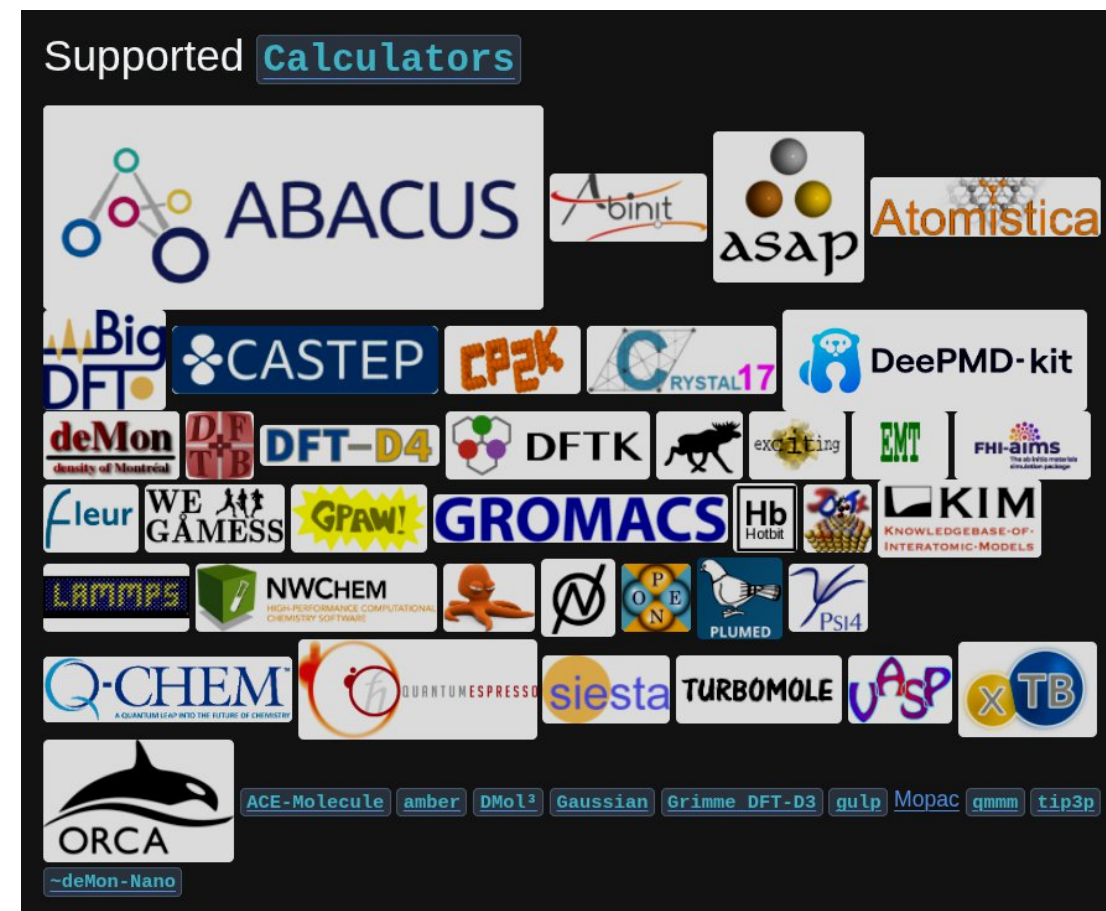
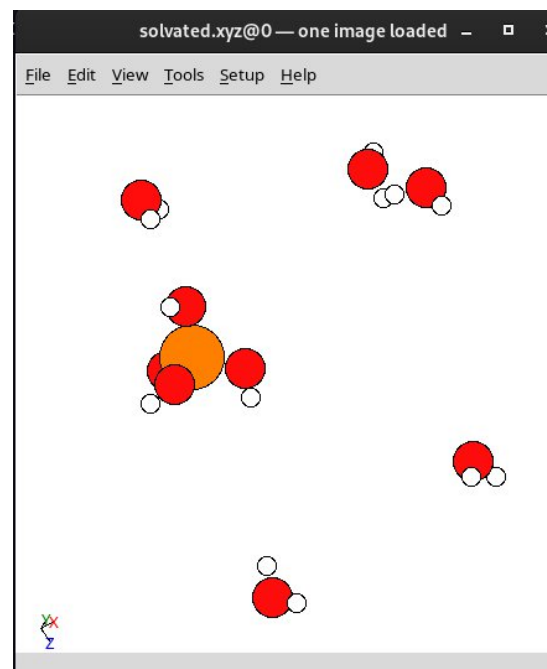


Introduction to Atomic Simulation Environment (ASE)



- Setting up, steering and analyzing: Python based
- Supports many Calculators/Software
- Easy to use (GUI, CLI)
- Open-source and cross-platform
- Data Analysis tools
- Movie making and Images

```
$ ase --help
$ ase sub-command --help
$ python -m module --help
```



Installation

- Linux: apt-get install python-ase
- Python: pip3 install --upgrade --user ase



Import required modules

```
from ase import Atoms
from ase.build import add_adsorbate, fcc111
from ase.calculators.emt import EMT
from ase.constraints import FixAtoms
from ase.optimize import QuasiNewton

h = 1.85
d = 1.10

slab = fcc111('Cu', size=(4, 4, 2), vacuum=10.0)
```

Example Script

```
from ase import Atoms
from ase.build import add_adsorbate, fcc111
from ase.calculators.emt import EMT
from ase.constraints import FixAtoms
from ase.optimize import QuasiNewton

h = 1.85
d = 1.10

slab = fcc111('Cu', size=(4, 4, 2), vacuum=10.0)

slab.calc = EMT()
e_slab = slab.get_potential_energy()

molecule = Atoms('N2', positions=[(0.0, 0.0, 0.0), (0.0, 0.0, d)])
molecule.calc = EMT()
e_N2 = molecule.get_potential_energy()

add_adsorbate(slab, molecule, h, 'ontop')
constraint = FixAtoms(mask=[a.symbol != 'N' for a in slab])
slab.set_constraint(constraint)
dyn = QuasiNewton(slab, trajectory='N2Cu.traj')
dyn.run(fmax=0.05)

print('Adsorption energy:', e_slab + e_N2 - slab.get_potential_energy())
```

create slab

calculate potential energy

add N2 on slab

run dynamics



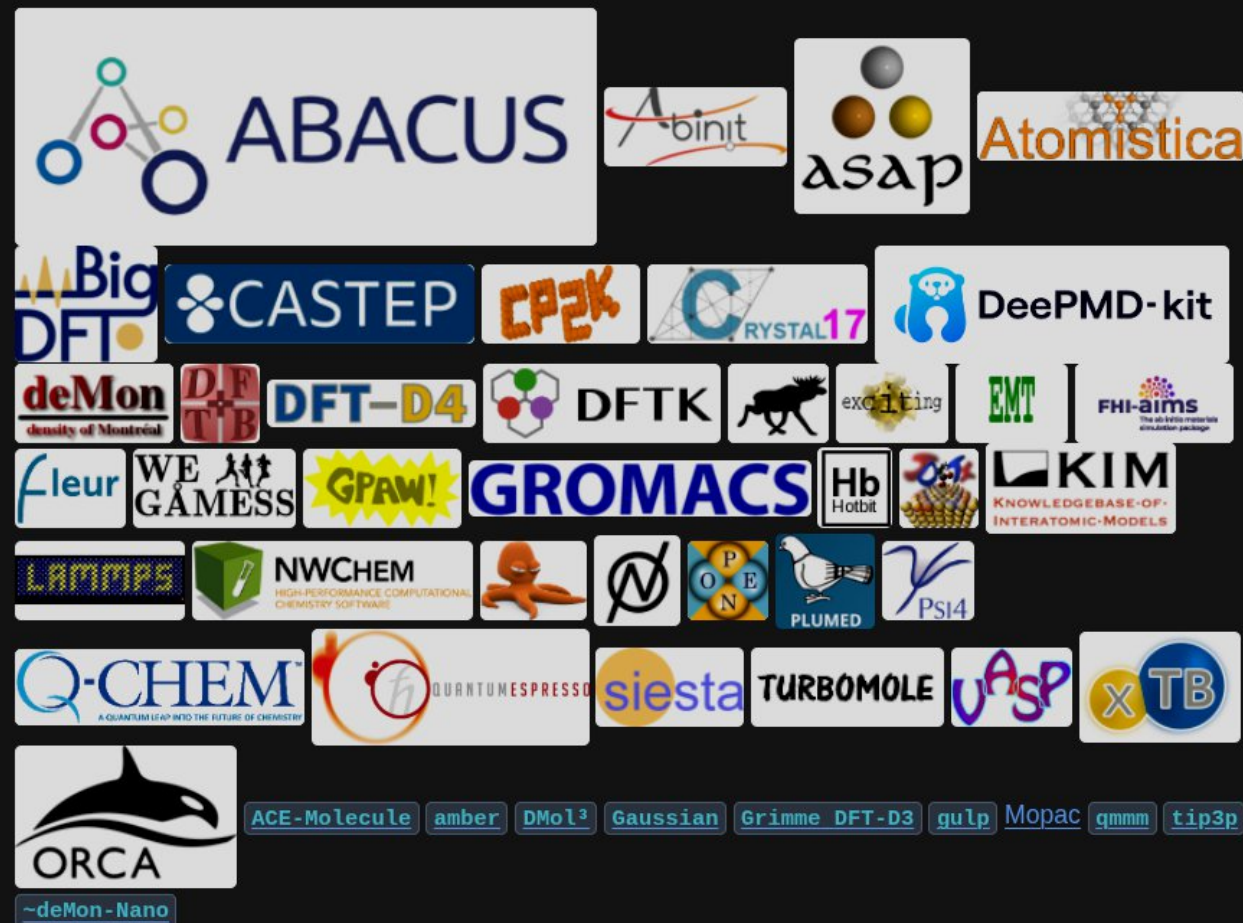
EMT(): simple calculator that can be used for quick demonstrations and tests

```
python N2Cu.py
```


Wide range of Calculators



Supported Calculators



```
[8]: #slab_vth_benzene.center(vacuum=5.0)
slab_vth_benzene.set_pbc([True, True, True])
slab_vth_benzene.calc = EMT() # Basic All-atom force field based

# calc = CP2K(label='benzene',
#             cutoff=250,
#             basis_set='SZV-MOLOPT-GTH',
#             pseudo_potential='auto',
#             charge=0,
#             xc='PBE',
#             print_level='LOW',
#             max_scf=20)
#slab_vth_benzene.calc = calc
```

Tutorials

ASE Tutorials

Atomization energy

Equation of state (EOS)

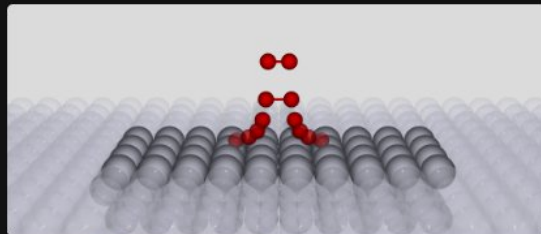
Dimensionality analysis

Finding lattice constants using
EOS and the stress tensor

Equilibrating an MD box of
acetonitrile

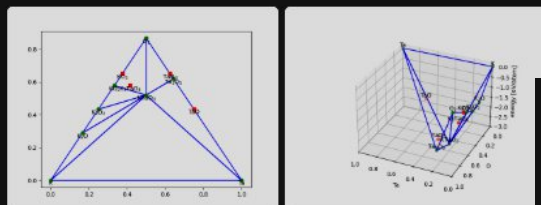
Molecular dynamics

Dissociation of oxygen on Pt(100)



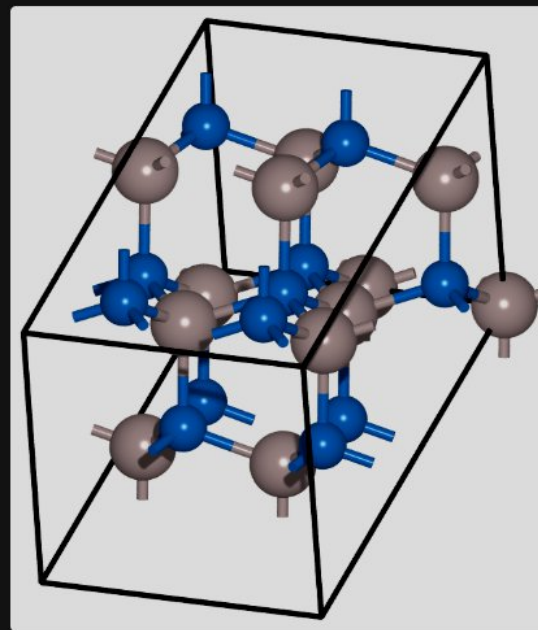
[ase.io](#) [o2pt100.py](#)

Phase diagrams



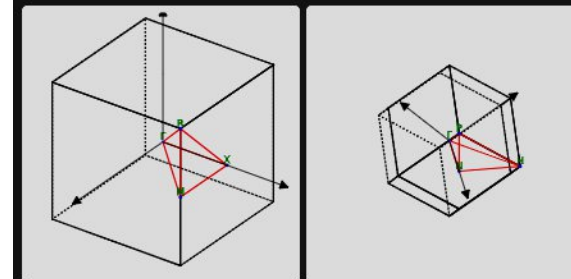
[ase.phasediagram](#) [ktao.py](#)

Fine tuning POV-Ray settings



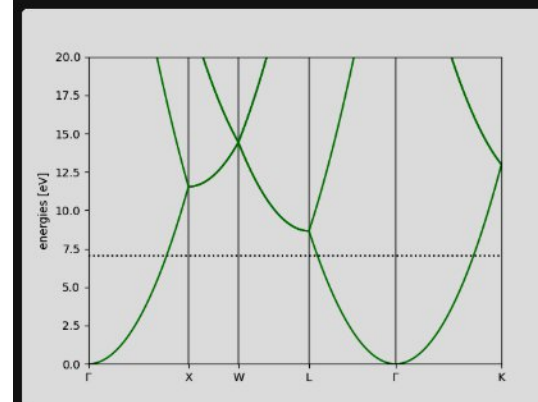
[tuning_povray_for_high_quality_images.py](#)

Brillouin zones



[ase.dft](#) [bz.py](#)

Band-structure



[ase.dft.kpoints](#) [bs.py](#)

<https://ase-lib.org/tutorials/tutorials.html>